



PHOTO FRAME GS 02

E-Ink Display 960x540px [LilyGo T5 E-paper Plus / ESP32 S3]

E-ink digital picture frame with SD + LittleFS image storage and remote management via FTP/Telnet (including HTTP(S) fetch), plus built-in file operations (list/copy/delete). Features dual-slot OTA firmware updates, protected Telnet authentication (session timeout + progressive lockout), and ultra-low-power deep sleep with RTC backup and scheduled wake-up hour for long battery life.

CONNECTION & ACCESS

WIFI AP MODE (Default)

SSID: PhotoFrameGS02
Password: 123456789
IP: 192.168.4.1
Gateway: 192.168.4.1
Subnet: 255.255.255.0

WIFI STA MODE

sta_ssid – WiFi name
sta_password – WiFi password
sta_enable – static IP
sta_ip/gateway/subnet/dns
Set: ap_enable = false

mDNS (optional)

mdns_enable = true
mdns_hostname = photoframegs02
Access: photoframegs02.local

TELNET ACCESS

Port: 23
User: admin
Pass: 123456789
Session: 1 hour

FTP ACCESS

Port: 21
User: admin
Pass: 123456789

TELNET COMMANDS

SYSTEM

help – command list
clear – clear screen
reboot – restart device
reset config – factory reset
exit – disconnect
logout – re-auth

INFO

meminfo – RAM/PSRAM
netinfo – IP/MAC/DNS
fsinfo – SD/LFS usage
sketchinfo – FW/sketch info
nvsinfo – NVS usage
batinfo – battery status
date – system time
date rtc – RTC time
date rtc set YYYY.MM.DD HH:MM:SS – set RTC
date rtc sync-from-ntp – RTC ← NTP
date rtc sync-to-system – SYS ← RTC
timestamp – Unix epoch

FILES

list – list active storage
list sd – list SD
list lfs – list LittleFS
cat <filename> – show file
copy sd lfs <file> – SD -> LFS
copy lfs sd <file> – LFS -> SD
delete sd <file> – delete from SD (confirm)
delete lfs <file> – delete from LFS (confirm)

CONFIG

config <key> – get value
config <key> <value> – set value

DOWNLOAD / UPDATE

fetch <url> [name] – HTTP/HTTPS, max 400KB, jpg/jpeg
fwupdate – update status
fwupdate verify – verify firmware
fwupdate run – apply update (confirm)
bootpart status[ota0]ota1 – boot slot info/set

FTP & CONFIGURATION

FILE STRUCTURE

/config.ini – settings
/images/ – photos (active storage: SD or LittleFS)

IMAGE REQUIREMENTS

Format: JPG/JPEG
Size: 960x540px
Max.: 400 KB per file
Tip: grayscale + contrast optimization
magic_pic01.jpg -contrast-space Gray -contrast-stretch 1.5x1.5 -ordered-dither ebl8,16 pic01.jpg

CONFIG.INI SECTIONS

[device] – name, version
[display] – jpg settings
[ap mode] – AP settings
[sta mode] – WiFi creds
[static ip] – IP settings
[ntp] – time server
[telnet] – telnet config
[ftp] – ftp config
[timer] – wake interval
[storage] – default FS + fallback

WAKE UP VALUES

1=Sec, **2**=Min, **3**=Hour, **4**=12h, **5**=Day, **6**=Week, **7**=Month, **wake_up_hour**=0..23 (used in Day/Week/Month modes)

BUTTONS

Maintenance/Wake (GPIO21): hold 3s during boot -> Maintenance mode
Factory Reset (GPIO48): hold 30s (Next image action if configured in your firmware flow)

OPERATING MODES

1 PHOTO FRAME MODE (Normal Operation)

- Automatically displays JPG images from the active **/images/** folder (SD or LittleFS)
- Uses storage fallback if the primary storage has no images
- Cycles images by configured wake mode: Seconds, Minutes, Hourly, Half-Day, Daily, Weekly, Monthly
- Supports wake_up_hour scheduling (0-23) for Daily/Weekly/Monthly wake modes
- Uses RTC-backed timing for accurate scheduled wake-ups
- Deep sleep between refresh cycles for battery conservation
- Low-battery detection with visual warning and power-safe behavior

2 MAINTENANCE MODE (Configuration & File Management)

Activation: Hold Maintenance Button (GPIO 21) for 3 seconds during boot

Features available:

- WiFi connectivity (AP or STA mode)
- Telnet server for remote configuration and file operations (list/copy/delete/fetch)
- Telnet auth protection (session timeout + lockout)
- FTP server for file upload/download
- mDNS support for easy hostname access
- OTA maintenance commands (firmware verify/run, boot slot control)

Exit: Hold Maintenance Button for 3 seconds to reboot

Factory Reset: Hold Reset Button (GPIO 48) for 30 seconds

3 LOW BATTERY MODE (Empty battery)

- Triggered when battery voltage drops below the low-battery threshold
- Shows battery status on display (voltage + percentage)
- Enters extended deep sleep to preserve remaining energy
- Wakes periodically for battery re-check and safe recovery handling

DESIGNED BY  FROM 

BUTTONS

NEXT IMAGE

RESET

MAIN-TENANCE

 USB-C CONNECTOR